KAVIRAJ K

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PROFILE SUMMARY

Embedded Engineer with experience in automotive electrical systems and embedded solutions, specializing in ATE (Automated Test Equipment) design and development for PCBA validation, BMS calibration, and ECU testing. Proficient in hardware-software integration, with hands-on experience in validation and verification of vehicle modules using real-time diagnostic and test systems.

TECHNICAL & KEY SKILLS

- **Core Competencies:** BMS Calibration, ECU Testing, ATE Development, PCBA Validation & Verification, Automotive Embedded Systems, Testing Methodologies (Basics + Learning)
- **Test & Hardware Expertise:**
  - ATE (Automated Test Equipment) Design & Development.
  - Hardware Integration & Troubleshooting.
  - Test Equipment Handling (DMM, Oscilloscope, Power Supplies, Bi-directional Power Supply, LED Analyzer, Battery Hi Tester (ACIR), DCIR Tester and Hipot Tester).
  - Panel Board Assembly & Wiring (Single Phase / Three Phase).
  - Electrical schematic and cable harness schematic reading & planning.
  - Equipment and component selection based on project requirements.
- **Protocols & Communication:** CAN, UART, SPI, I2C, and K-Line.
- **Programming Languages:** Embedded C, C, C++ (Basics), Python (Basics + Learning).
- **Microcontrollers:** PIC16FXXX, STM32FXXX.
- **Development Tools & IDEs:** MPLAB X IDE, Keil uVision, STM32 Cube IDE, Code Blocks.
- **Communication Modules:** Bluetooth (HC-05), GSM.
- **Embedded Peripherals:** ADC, Timers, Interrupts, RTC (DS1307).
- **Documentation:** BOM, LLD Documentation.
- **Productivity Tools:** MS Office (Word, Excel, PowerPoint), Solidworks (Basics + Learning).
- **Operating Systems:** Windows (up to 11), Linux (Basics).
- **Soft Skills:** Team Collaboration, Customer Communication.

EXPERIENCE

Rennova Innovation & Solutions Pvt. Ltd., Hosur. **December 2023 – Present**

Embedded Engineer - Automotive Testing

- Led end-to-end ATE development lifecycle, including requirement gathering, SOR review and customer discussions.
- Collaborated with customers and cross-functional teams (hardware, mechanical, software) to define requirements, resolve issues, and ensure timely project execution.
- Coordinated with purchase, production, and suppliers for component selection, procurement, and system integration.
- Prepared BOM and Low-Level Design (LLD), and conducted bench-level testing aligned with customer requirements.
- Designed and implemented automated test sequences, reducing manual testing effort and improving validation efficiency.
- Performed system-level validation of PCBAs and vehicle modules, ensuring compliance with functional and diagnostic requirements.

- Utilized test equipment (Programmable power supplies, DMMs, Oscilloscopes, LED Analyzers) for validation and troubleshooting.
- Implemented communication interfaces using CAN, K-Line, UART, SPI, and I2C for ATE and DUT interaction.
- Handled ATE installation, customer demonstrations, and end-user training.

Key Projects Handled:

- **BMS Shunt Calibration Testing:** Performed current shunt calibration using bidirectional power supply across charge/discharge cycles; monitored parameters like cell voltage, SOC, bus and auxiliary voltage via hardware and CAN communication.
- **Climate Control Module (CCM):** Tested the functionality of the PCBA before and after firmware flashing by verifying voltage and resistance at multiple checkpoints. Ensured accurate data transmission over CAN (UDS) and LIN diagnostic protocols, checking for data integrity and confirming absence of data loss or negative responses (NR).
- **Vehicle Sunroof Control Unit:** Tested PCBA functionality to verify correct firmware flashing then measuring voltage, resistance, and current under specific conditions to ensure proper operation. Validated reliable data transmission over the K-Line protocol, ensuring data integrity using CRC32 checksum and confirming absence of data loss.
- **Forklift Functionality Testing:** Tested forklift systems for correct CAN responses at specified IDs when switches, sensors, and other inputs were triggered. Verified functionality of lights and buzzers using lux meters and sound sensors. Conducted electrical validation across chassis, cluster, and engine systems by measuring voltage, current, and resistance.
- **Wiring Harness Testing and Net list Creation (Currently Handling):** Tested the vehicle wiring harness for continuity, short circuits, insulation resistance, and data loss to ensure proper functionality and electrical integrity.
- **Vehicle LED Functionality Testing:** Tested vehicle tail lamps by transmitting corresponding CAN signals and verifying functionality through communication with master-slave PCBAs; evaluated LED intensity using an LED analyzer.
- **Battery Module and Battery Pack Testing & Validation:** Performed ACIR, DCIR, cell voltage, temperature, and Hi-Pot testing to evaluate battery health, internal resistance, performance, insulation integrity, and overall product quality. Supported battery module validation, battery pack assembly, functional verification, troubleshooting, and quality assurance activities.

Previous Experience

- Previously worked in automation and electrical domains, including panel-related work and associated tasks.

EDUCATIONAL PROFILE

- BE (Electrical & Electronics Engineering) –
  Ranganathan Engineering College, Anna University

CERTIFICATIONS

- Diploma in Embedded Systems (6 Months) – Manfree Technologies, Coimbatore.

PERSONAL PROFILE

Current Location : Hosur
Languages Known : Tamil, English (To Read, Write and Speak)